

DETAILED DESCRIPTION OF THE INVENTION — POWER SYSTEMS, CHARGING ARCHITECTURES, AND ENERGY MANAGEMENT

The intraluminal physiologic monitoring device may incorporate a variety of power supply and energy management architectures configured to support reliable operation within compact form factors and biologic environments.

In various embodiments, electrical power may be supplied by rechargeable batteries, disposable cells, thin-film energy storage devices, solid-state batteries, flexible batteries, supercapacitors, or hybrid energy storage systems. Battery placement may occur within distal, proximal, centralized, or distributed regions of the device depending on design constraints and performance requirements.

Charging architectures may include inductive wireless charging, resonant magnetic coupling, capacitive charging, magnetic contact charging, sealed pogo-pin interfaces, conductive docking interfaces, or external charging cartridges. Charging systems may be configured to maintain waterproof integrity while enabling convenient user interaction.

Certain embodiments may employ energy harvesting techniques including motion-based energy generation, thermal gradient harvesting, radio-frequency energy capture, piezoelectric conversion, or biomechanical energy recovery systems capable of supplementing stored energy.

Power management circuitry may dynamically regulate sensor activation, processor operation, wireless transmission intervals, and standby states to optimize operational lifetime. Adaptive energy management algorithms may modify power consumption based on detected usage patterns, physiologic events, or signal quality requirements.

Low-power operational modes may allow intermittent sensing, scheduled measurement windows, or event-triggered activation designed to extend usable duration between charging cycles. Sleep states, deep standby modes, or sensor duty cycling may further reduce energy consumption.

Thermal management considerations may influence power architecture design to minimize heat generation within biologic environments. Power regulation components may include thermal monitoring, load balancing, or automatic power throttling mechanisms.

Modular power systems may allow removable battery modules, externalized power packs, reusable electronics cores, or disposable sensing shells powered through detachable energy sources. Such configurations may support both consumer and clinical deployment models.

Future embodiments may incorporate emerging energy storage technologies, wireless power ecosystems, or miniaturized autonomous power systems capable of achieving substantially similar operational outcomes.

All power supply, charging, and energy management approaches capable of supporting intraluminal physiologic monitoring functionality are intended to fall within the scope of the present disclosure.